

Name of the Student	K. Lokesh Chandra
Reg. No	192425358
GitHub Link	https://github.com/Lokesh-123-06/Deep-Learning

SIMATS ENGINEERING

CO2 - ASSESSMENT TOOL 2

Scenario-Based Questions

COURSE NAME: Deep Learning

COURSE CODE: MLA04

COURSE OUTCOME COVERED

CO 2:Students will be able to apply supervised learning algorithms such as linear regression, classification models, decision trees, neural networks, support vector machines, and ensemble methods to solve real-world prediction and classification problems. They will gain the ability to select appropriate models, train them using datasets, and evaluate their performance. Students will also understand the strengths and limitations of different supervised learning approaches.(BTL-4)

CO Weightage : 15%

Assessment Tool Weightage: 40%

Duration: 90 minutes

Marks: 25

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

K. Lokesh Chandra

Name of the student:

K. Lokesh Chandra

Scenario-Based Questions

CO-AT2

QUESTIONS: 05

Total Marks:25

1. Unsupervised Training of Neural Networks, Auto Encoders

A hospital wants to identify abnormal ECG patterns from thousands of unlabeled records. Which deep learning technique from the syllabus would you recommend? Justify your choice and explain the workflow.

2. Auto Encoders, Representation Learning

A retail company wants to reduce the dimensionality of customer purchase data before performing customer segmentation. Explain how an autoencoder can be used to solve this problem.

3. Width and Depth of Neural Networks, Activation Functions

A facial recognition system performs poorly when trained using a shallow neural network. Recommend architectural changes involving network depth and activation functions.

4. Restricted Boltzmann Machines (RBMs), Unsupervised Training of Neural Networks

A streaming service wants to recommend movies to users based on their viewing history without using labeled data. Explain how RBMs can be applied to build the recommendation engine.

5. Deep Learning Applications, Representation Learning

A manufacturing company wants to detect defective products from sensor readings collected every second. Design a deep learning solution using concepts from representation learning and neural networks.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships	Shows correct relationships with minor	Relationships are weak, unclear, or partially	No meaningful relationships

		between concepts	gaps	incorrect	established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

ANSWERS:

1. Unsupervised Training of Neural Networks, Autoencoders

Answer:

The most suitable deep learning technique is an **Autoencoder**, which is an **unsupervised neural network** used for anomaly detection.

Justification

- ECG records are **unlabeled**, so supervised learning cannot be applied.
- Autoencoders learn the **normal patterns** of ECG signals by compressing and reconstructing the input.
- When an abnormal ECG is given as input, the reconstruction error becomes high because the model has not learned abnormal patterns.
- Records with high reconstruction error are identified as abnormal.

Workflow

1. Collect unlabeled ECG data.
2. Preprocess the ECG signals (remove noise and normalize).
3. Train the autoencoder using normal ECG patterns.
4. Encode the input into a low-dimensional representation.
5. Decode the representation to reconstruct the ECG.

6. Calculate the reconstruction error.
7. If the error is greater than a threshold, classify it as an abnormal ECG; otherwise, classify it as normal.

Advantages

- Works without labeled data.
- Detects unknown abnormalities.
- Reduces manual inspection by doctors.

2. Autoencoders, Representation Learning

Answer:

An **Autoencoder** can reduce the dimensionality of customer purchase data by learning a compact representation of the original dataset.

Working

1. Collect customer purchase data.
2. Normalize and preprocess the data.
3. Feed the data into the encoder.
4. The encoder compresses the high-dimensional features into a smaller latent representation.
5. The decoder reconstructs the original data.
6. The compressed features are extracted from the bottleneck layer.
7. Use these features for customer segmentation using clustering algorithms such as K-Means.

Benefits

- Removes redundant information.
- Preserves important customer behavior patterns.
- Reduces computation time.
- Improves clustering accuracy.
- Handles large datasets efficiently.

3. Width and Depth of Neural Networks, Activation Functions

Answer:

To improve facial recognition performance, the neural network should be made **deeper** and use better **activation functions**.

Recommended Changes

Increase Network Depth

- Add more hidden layers.
- Deep networks learn low-level, medium-level, and high-level facial features.
- Improves feature extraction and recognition accuracy.

Use Better Activation Functions

- Replace Sigmoid with **ReLU** because it reduces the vanishing gradient problem.
- Use **Leaky ReLU** if dead neurons occur.
- Use **Softmax** in the output layer for face classification.

Expected Outcome

- Better learning of complex facial features.
- Faster training.
- Higher recognition accuracy.
- Better generalization for different lighting conditions, angles, and facial expressions.

4. Restricted Boltzmann Machines (RBMs), Unsupervised Training of Neural Networks

Answer:

A **Restricted Boltzmann Machine (RBM)** can be used to build a movie recommendation system using users' viewing history without labeled data.

Working

1. Create a user–movie interaction matrix.
2. Each visible node represents a movie.
3. Hidden nodes learn user preferences automatically.
4. Train the RBM using unsupervised learning.
5. The RBM identifies hidden relationships between users and movies.
6. Predict movies that users are likely to watch.
7. Recommend movies with the highest predicted ratings.

Advantages

- Learns hidden viewing patterns.

- Works without labeled datasets.
- Handles sparse data effectively.
- Provides personalized movie recommendations.
- Improves user satisfaction.

5. Deep Learning Applications, Representation Learning

Answer:

A deep learning system using **representation learning** and **neural networks** can detect defective products from continuous sensor data.

Proposed Solution

Step 1: Data Collection

Collect sensor readings from manufacturing equipment every second.

Step 2: Data Preprocessing

- Remove noise.
- Normalize values.
- Handle missing data.

Step 3: Representation Learning

Use an **Autoencoder** to learn compact representations of normal sensor behavior.

Step 4: Neural Network Classification

Feed the learned features into a Deep Neural Network to classify products as:

- Normal
- Defective

Step 5: Decision

If reconstruction error or classification probability exceeds a threshold, mark the product as defective and generate an alert.

Benefits

- Detects defects in real time.
- Reduces manual inspection.
- Improves product quality.
- Minimizes production losses.
- Supports predictive maintenance.

